

DUNKIRK LH, NY, USA

WMO: 994250

Lat: 42.4939N

Lon: 79.3540W

Elev: 600

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.4	(c) 10.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 56.6	(k) 33.7	(l) 50.5	(m) 32.3	(n) 15.3	(o) 220	(p) 0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.9	(c) 81.8	(d) N/A	(e) 79.9	(f) N/A	(g) 78.2	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 9.0	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 43.5	(b) 36.2	(c) 31.3	(e) DB	(f) 2.8	(g) 87.4	(h) 6.4	(i) 2.1	(j) -1.8	(k) 88.9	(l) -5.5	(m) 90.2	(n) -9.1	(o) 91.4	(p) -13.8	(q) 92.9	(r) N/A
			(e) WB	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A	(r) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	27.8	28.3	33.9	44.8	55.8	66.0	70.9	70.2	64.4	53.9	43.6	34.5
	DBStd	17.39	10.83	9.89	9.79	8.49	7.67	6.21	4.46	4.28	6.17	7.32	8.46	8.59
	HDD50	2794	691	609	508	203	28	0	0	0	41	227	487	
	HDD65	6150	1153	1027	963	606	303	62	5	7	84	352	641	947
	CDD50	2660	4	2	10	48	206	480	647	627	431	162	37	6
	CDD65	538	0	0	0	1	17	92	187	169	64	8	0	0
	CDH74	1911	0	0	1	11	49	288	726	595	222	19	0	0
	CDH80	158	0	0	0	1	5	28	55	44	25	0	0	0
(14) Wind	WSAvg	12.8	17.1	15.5	14.2	14.1	11.9	9.4	9.3	8.8	10.0	12.6	15.3	16.2
(15) Precipitation	PrecAvg	42.1	2.6	2.0	2.6	3.4	3.4	3.7	3.8	3.9	4.9	4.1	4.4	3.3
	PrecMax	58.1	5.6	5.2	5.4	6.4	6.8	9.8	9.2	9.0	11.9	7.4	12.1	7.0
	PrecMin	31.4	0.9	0.4	0.6	0.9	0.9	1.1	1.1	0.9	2.1	0.9	1.5	1.3
	PrecStd	6.3	1.1	1.1	1.1	1.3	1.5	1.8	1.9	1.7	2.2	1.4	2.0	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.9	61.8	67.7	75.4	79.4	83.5	84.2	83.9	83.3	76.7	67.9	63.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	55.4	52.8	60.7	68.8	74.4	79.7	81.4	81.0	79.1	72.1	63.8	55.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	47.8	46.1	54.7	63.3	71.0	77.3	79.6	79.1	76.5	68.3	60.2	50.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	41.7	40.5	47.7	58.3	67.6	75.0	77.9	77.4	73.8	64.8	56.3	46.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	11.1	11.7	12.9	12.1	10.5	9.9	10.3	11.3	10.7	9.9	9.3
		MCDBR	16.8	19.0	21.8	23.3	17.3	13.5	11.2	11.5	13.7	15.0	15.2	14.9
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.306	0.342	0.337	0.375	0.405	0.437	0.439	0.429	0.380	0.362	0.316	0.304
	taud		2.393	2.257	2.366	2.321	2.285	2.232	2.249	2.269	2.377	2.407	2.506	2.459
	Ebn at Noon		264	269	287	283	276	266	264	263	270	260	260	255
	Edh at Noon		25	34	34	39	41	44	43	41	34	29	23	22
(44) All-Sky Solar Radiation	RadAvg		406	671	1006	1398	1722	1875	1889	1677	1320	804	493	310
	RadStd		37	67	131	148	175	141	144	94	123	82	76	44

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page